

AKHIL M M

FULL STACK DEVELOPER

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Summary

Full-stack developer experienced in building web applications using TypeScript, Node.js, React, MongoDB, and Redis. Developed production-style projects including a driver-on-demand platform, AI-powered real-time chat application, and e-commerce platform with payment integrations, real-time communication, and cloud deployment. Interested in full-stack development, system design, and building reliable, maintainable software.

Technical Skills

Languages: TypeScript, JavaScript (ES6+)

Backend: Node.js, Express.js, REST APIs, Socket.IO, JWT Authentication

Frontend: React.js, Redux Toolkit, Next.js, Tailwind CSS, Material UI, HTML, CSS

Database & Caching: MongoDB, Redis, MongoDB Atlas Vector Search

Cloud & DevOps: AWS EC2, AWS S3, Google Cloud Platform (GCP), Docker, NGINX, PM2, Vercel

Tools & Integrations: Git, GitHub, Postman, Cloudinary, Stripe, Razorpay, WebRTC, Google Maps API

Projects

Sarathi - Driver-On-Demand Platform

Live — GitHub

- Architected the backend from scratch using TypeScript, Clean Architecture, and Tsyringe dependency injection while following SOLID principles for maintainable and scalable codebases.
- Built a real-time driver matchmaking system using Socket.IO and Google Maps API to broadcast ride requests and assign nearby approved drivers based on road-distance calculations.
- Developed peer-to-peer chat and video calling features using WebRTC and Socket.IO, with Redis-backed state management supporting real-time messaging, media sharing, and user-driver communication.
- Integrated Stripe Connected Accounts to automate driver onboarding, split-payment processing, and secure transaction handling within the booking workflow.
- Deployed the platform on AWS EC2 using PM2 and NGINX, hosted the frontend on Vercel, and integrated Cloudinary for secure media storage and delivery.

AI-Powered Real-Time Group Chat Platform

Live — GitHub

- Integrated Google Gemini API with a custom Retrieval-Augmented Generation (RAG) pipeline using MongoDB Atlas Vector Search and Gemini embeddings to deliver context-aware AI responses.
- Architected the backend using TypeScript and Clean Architecture with modular Use Cases, Repositories, Services, and Controllers for maintainability and scalability.
- Developed real-time messaging features using Socket.IO, including AI-assisted conversations, read receipts, intelligent message grouping, media sharing, and event-driven communication.
- Implemented high-performance virtualized message rendering using TanStack Virtual, infinite scrolling, and minimized re-renders to efficiently handle large chat histories.
- Implemented JWT access/refresh token rotation, Google OAuth 2.0 authentication, AWS S3 presigned uploads, and deployed backend services on Google Cloud Platform (GCP).

EYEGAZE - Full-Stack E-Commerce Platform

Live — GitHub

- Built a full-stack eyewear e-commerce platform with secure authentication, product catalog management, inventory tracking, and order processing workflows.
- Integrated Razorpay payment gateway to support secure checkout and online payment processing.
- Implemented product, category, inventory, offer, and coupon management modules to streamline store administration and business operations.
- Developed an administrative dashboard using Chart.js to visualize revenue, sales trends, user growth, and business analytics.

Experience

Venkys Group

Marketing Assistant

2022 – 2024

Kerala, India

- Supported marketing and customer engagement activities while coordinating with internal teams and assisting in day-to-day operational processes.

Education

Prototype	2024 – 2026
<i>Professional Training - Full Stack Development (MERN Stack)</i>	Kochi, Kerala
University of Calicut	2019 – 2022
<i>B.Sc. Mathematics with Complementary Studies in Computer Science</i>	Kerala, India